

Logisoft HireOS

AI Recruitment Platform — Demo Brief for CEO & Director

One line: HireOS reads the job, screens resumes, interviews candidates with AI, tests their skills, and gives the recruiter clear evidence — **AI assists, humans decide.**

Tagline: "Intelligent hiring. Human decisions." · 100% self-hosted · Prepared 29 Sep 2026

What HireOS does

1. Manage hiring

Jobs, candidates, resumes, pipeline board, full activity history.

2. AI screening

Matches resume to job. Gives score, strengths, missing skills, concerns.

3. AI interview

AI asks questions by text or voice, adapts follow-ups, records answers.

4. Skill tests

Coding (Python / JavaScript) and SQL tests run in a locked sandbox.

5. Integrity checks

Tab switch, paste, camera, optional phone as 2nd camera — flagged, not judged.

6. Evidence report

Per-answer reasoning, overall summary, skill-by-skill evidence.

Three messages to land

1. **Saves recruiter time** — screening, interviewing and first-round testing run by the system.
2. **Fair and explainable** — every AI score shows its reasoning; nothing is a black box.
3. **Our data stays with us** — no cloud AI, no per-candidate fees, runs on our own server.

Golden rule of the product: AI never selects or rejects anyone and never moves a candidate's stage. Only a recruiter can — and Select / Reject needs a written note.

How it is built

Frontend (what users see)
Next.js 14 + React, TypeScript, Tailwind. Recruiter dashboard, careers page, candidate portal, interview room.

Backend (logic & APIs)
Next.js API (live today). A Django + Celery backend is also built and tested, kept switched off for now.

Database
PostgreSQL with pgvector (for smart “search by meaning”). Files (resumes, recordings) on our own disk.

AI brain
Qwen 2.5 (7B) open-source model, running locally through Ollama. Nomic model for smart search.

Voice
Whisper (speech → text) and Piper (text → voice). Both run locally.

Camera & code safety
Google MediaPipe in the browser for camera checks. Docker sandbox (no internet) to run candidate code.

Everything runs on **our machine / server**. No OpenAI, no cloud database, no cloud storage. Candidate data never leaves the company.

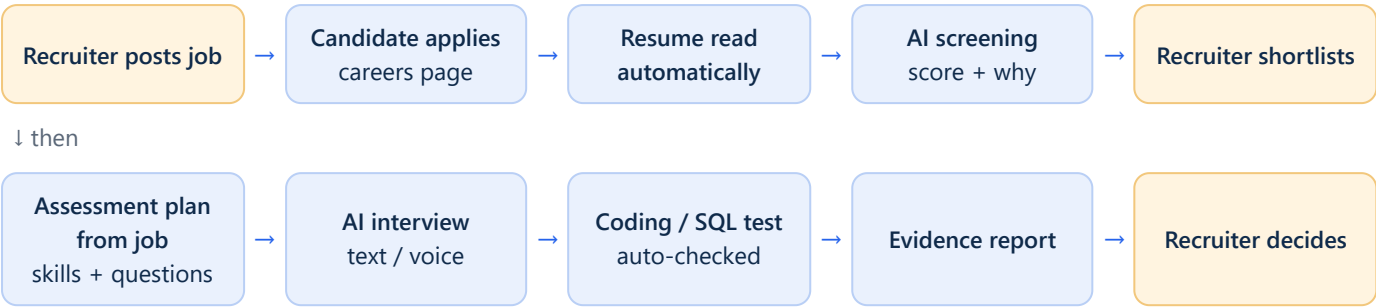
Third-party tools used (all free / open-source, run locally)

Tool	What it does for us
Ollama + Qwen 2.5 7B	The AI that screens, generates questions, evaluates answers
Nomic embed	Talent-pool search by meaning (e.g. “React dev with fintech”)
Whisper / Piper	Voice interview: listens to candidate, speaks questions
MediaPipe (Google)	Face / object detection for integrity signals, inside the browser
PostgreSQL + pgvector	Main database
Docker	Runs the database, AI, voice service and the code sandbox
Redis + Celery	Background job queue for the Django backend (ready, not switched on)
Caddy	Local HTTPS so a phone can join as the 2nd camera (QR code)
PDF / Word readers	Read uploaded resumes (pdf-parse, mammoth)
Email (SMTP)	Email templates are built; needs our company mail server connected

User roles (each sees only what they should)

Super Admin · HR Admin · Recruiter · Hiring Manager · Interviewer · Candidate

How it works — end to end



Blue = system does it · Orange = person does it

Hiring stages

Applied → Screening → Shortlisted → Assessment → AI Interview → Tech Interview → HR Interview → **Selected / Rejected**

Step-by-step process

#	Step	What happens
1	Create job	Recruiter adds title, description, skills, experience.
2	Candidates apply	Public careers page or recruiter uploads resume. Resume text read automatically.
3	AI screening	One click (or "Screen all"). Score, strengths, gaps, concerns, suggestion.
4	Shortlist	Recruiter moves good candidates forward on the pipeline board.
5	Assessment plan	System reads the job and builds a plan: skills to test, questions, difficulty, which practical test.
6	Send assessment	Recruiter clicks "Create from blueprint" / "Assign", then shares one link with the candidate.
7	Candidate takes it	AI interview (about 30 min) + coding / SQL test. Integrity signals recorded if enabled.
8	Auto evaluation	AI scores each answer with reasoning; code/SQL checked against hidden tests.
9	Review	Recruiter sees report, transcript, skill evidence, integrity signals.
10	Decision	Recruiter moves stage / selects / rejects with a note. Logged in history.

How automatic is it?

System does automatically

- Reads resumes (PDF / Word)
- Screening score with reasons (after one click)
- Assessment plan from the job description
- Interview questions (AI-worded, safety-checked)
- Conducts the interview, asks follow-ups
- Voice: speaks questions, transcribes answers
- Scores each answer + overall report with reasoning
- Runs and checks coding / SQL answers
- Records integrity signals with timestamps
- Keeps a full audit history of every action

Recruiter does manually

- Create the job
- Click “Run screening” / “Screen all”
- Shortlist and move stages
- Create interview / assign tests
- Copy and send the candidate link
(auto-email once mail server is connected)
- Review report and evidence
- Final Select / Reject with a note

In short: the heavy work (reading, screening, interviewing, testing, scoring) is automatic. The **decisions** stay human — by design.

Timing on the current machine (AI runs on CPU)

Task	Approx. time
Resume screening	1–2 minutes per candidate
Interview from assessment plan	Instant (AI wording improves in background)
Interview for candidate	About 30 minutes (set by recruiter)
Final interview report	A few minutes after interview ends
Coding / SQL check	A few seconds

A GPU server will make AI steps much faster. No per-candidate AI cost either way.

Demo walkthrough (about 15–20 min)

Log in as **recruiter@local.dev** at <http://localhost:3000/login> (password shared separately). Use existing demo candidates.

	Show	Say
1	Dashboard — numbers, pipeline summary	<i>"One place for all hiring: jobs, candidates, interviews."</i>
2	Jobs → Senior Full Stack Engineer	<i>"Every job holds its description, skills and candidates."</i>
3	Careers page (/careers) — open a job	<i>"Candidates apply here; the resume is read automatically."</i>
4	Pipeline board	<i>"Stages from Applied to Selected. Only people move candidates."</i>
5	Arjun Mehta → AI Screening card	<i>"AI gives a match score, strengths, gaps, concerns — and why."</i>
6	Job → Assessment page (blueprint)	<i>"From the job, the system plans what to test and how hard."</i>
7	Arjun's interview report (or Kavya Menon's)	<i>"Full transcript, every answer scored with reasoning, overall summary."</i>
8	Interview Links page	<i>"Each candidate gets a private link. We can see who started or finished."</i>
9	Candidate page → Assessment section	<i>"One link gives the candidate interview + coding/SQL test. We see progress and evidence per skill."</i>
10	Recruiter Decision panel (don't click)	<i>"AI never decides. Select or Reject needs a human note."</i>
11	Talent Pool search	<i>"Search past candidates by meaning, not just keywords."</i>
12	Analytics + Settings (templates, users)	<i>"Hiring metrics, email templates, role-based access."</i>

Optional live moments (only if time allows)

- **Live AI interview:** open Aisha Rahman's scheduled link from Interview Links; answer 2–3 questions.
- **Live assessment:** Meera Iyer (Senior Full Stack) → Assessment section → "Create from blueprint" + Assign Coding → Issue link → open the link → candidate sees one page with Interview + Coding.
- **Live SQL test:** Rohan Desai (People Analytics) → Assign SQL → open link → run a query → results appear for recruiter.
- **Phone 2nd camera:** interview in Enhanced mode shows a QR code; the phone joins as a side camera.

Live actions create real records on that candidate and AI steps take 1–2 min on this machine. If short on time, stick to steps 1–12 — the demo candidates already show finished results.

What we have implemented

Module	What is ready	Status
Foundation	Login, 6 roles, organisation & departments, secure sessions	Done
ATS core	Jobs, candidates, resume upload & reading, pipeline board, activity history	Done
Careers & portal	Public job page, apply form, candidate login & profile	Done
AI screening	Score, strengths, gaps, concerns, suggestion — with reasoning	Done
AI interview	Text & voice, adaptive follow-ups, timer, recording	Done
Integrity (proctoring)	Tab / paste / fullscreen, camera checks, phone 2nd camera — signals only	Done
Evaluation report	Per-answer and overall AI evaluation with reasoning	Done
Talent pool, templates, analytics	Smart search, email templates, dashboards	Done
Assessment Engine	Job → skill plan, question plan, difficulty, rubric	Done
AI question writer	AI rewrites questions, checked by guardrails, safe fallback	Done
Practical tests	Coding (Python, JavaScript) + SQL in locked sandbox, auto-checked	Done
Unified assessment	One candidate link, progress, skill-by-skill evidence for recruiter	Done
Django backend	Built and tested; ready to switch on stage by stage	Ready, off

Quality check: 298 unit tests, 159 security/isolation tests, 43 sandbox security tests, 168 backend tests — all passing.

What is pending

Item	Why / what's needed	Type
Email sending	Connect company mail server (SMTP) for auto-emails	Setup
Production server	Install on company server (GPU recommended), HTTPS, backups	Setup
Sandbox small fix	Two identical test submissions in the same second — fix ready for approval	Small fix
Switch on Django backend	Move work to the new backend step by step	Planned
Database history cleanup	Technical housekeeping of database version history	Planned
Phone-camera “device use” detection	Accuracy tuning, deliberately paused	Deferred
Next phase (V4)	AI skill scoring from all evidence, smarter adaptive interview	Next phase
More tests & languages	More coding / SQL tasks, more languages, stronger sandbox	Next phase
Code review	Latest assessment modules to be reviewed and saved to repository	Housekeeping

Likely questions — short answers

Does the AI hire or reject people?

No. It only suggests. A recruiter makes every decision and must write a note.

Which AI do we use?

Qwen 2.5 — an open-source model running on our own machine via Ollama. Not ChatGPT, not cloud.

Where is candidate data stored?

On our own server — database and files. Nothing is sent outside.

What does it cost to run?

No per-candidate or per-API fees. Main cost is the server (GPU recommended).

Can candidates cheat?

We track tab switches, paste, camera and an optional phone side-camera. These are flags with time stamps for a human to review — they never change the AI score.

Is the AI fair / explainable?

Every score stores its reasoning. Recruiter sees why, not just a number.

Is it secure?

Role-based access, company data isolation, secure login, full audit trail. Candidate code runs in a locked sandbox with no internet.

How much is automatic?

Reading, screening, interviewing, testing and scoring. People create jobs, send links and decide.

Is it ready to use?

Ready for a pilot on our local server. Pending: mail server, production server setup, a few small items listed on the previous page.

Before the demo — checklist

- Say **“lets go”** to start everything (database, AI, backend, website).
- Start the code sandbox if showing a live test: `npm run sandbox:runner`.
- Open <http://localhost:3000/login> — log in as recruiter once to warm up.
- Open Arjun Mehta and Kavya Menon pages beforehand (finished results ready).
- Keep this PDF open on a second screen.

Logisoft HireOS — Intelligent hiring. Human decisions.